



PYTHON BOOTCAMP

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MODULE & LIBRARIES



Module:

- A file containing code (functions, classes, variables, etc.) that can be imported and used in other Python programs.

Libraries:

- A collection of modules and packages that provide specific functionality.

MODULE & LIBRARIES



Module:

1. Create math_utils.py

```
1 def add(a, b):
2     """Add two numbers"""
3     return a + b
4
5 def multiply(a, b):
6     """Multiply two numbers"""
7     return a * b
8
9 def factorial(n):
10    """Calculate factorial of n"""
11    if n <= 1:
12        return 1
13    return n * factorial(n - 1)
14
15 PI = 3.14159
16
17 class Calculator:
18     def __init__(self):
19         self.history = []
20
21     def calculate(self, operation, a, b):
22         if operation == "add":
23             result = add(a, b)
24         elif operation == "multiply":
25             result = multiply(a, b)
26         else:
27             result = None
28
29         self.history.append(f"{operation}({a}, {b}) = {result}")
30     return result
```

MODULE & LIBRARIES



Module:

2. Import functions or class from math_utils.py

```
3  from math_utils import add, multiply, factorial, PI, Calculator
4
5  result = add(5, 3)
6  print(f"Addition Result: {result}")
```

MODULE & LIBRARIES



Libraries:

```
21 import os
22 import sys
23 import datetime
24 import random
25
26 sys.path.append(os.path.dirname(os.path.dirname(os.path.abspath(__file__)))) # Adjust path for imports to root directory
27
28 now = datetime.datetime.now()
29 today = datetime.date.today()
30 formatted_date = now.strftime("%Y-%m-%d %H:%M:%S")
31
32 print(f"Now date: {now}")
33 print(f"Today's Date: {today}")
34 print(f"Current Date and Time: {formatted_date}")
35
36 random_number = random.randint(1, 100)
37 random_choice = random.choice(['apple', 'banana', 'orange'])
38 numbers = [1, 2, 3, 4, 5]
39 random.shuffle(numbers)
40
41 print(f"Random Number: {random_number}")
42 print(f"Random Choice: {random_choice}")
43 print(f"Shuffled List: {numbers}")
```